Study Session 3: Correlation, Regression, and Storytelling

■ **Objective:** Learn how variables relate using correlation and regression. Practice interpreting results and writing data stories.

1■■ Setup & Launch

- Activate your environment: conda activate ds
- Launch Jupyter Lab: jupyter lab
- Open your notebook: notebooks/Study_Session_3.ipynb

2■■ Explore Correlation

- Import seaborn and matplotlib
- Run: sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
- Interpret strong/weak correlations and signs (+ / -)

3■■ Simple Linear Regression

- Use scikit-learn: LinearRegression()
- Predict tip_usd from bill_total_usd
- Print slope, intercept, R^2 score
- Interpret meaning of R^2 and slope

4■■ Visualize Relationships

- Use plotly.express.scatter with trendline='ols'
- Add proper title, axis labels, and color by gender
- Save your chart to reports/figures/

5■■ Write Your Data Story

- In Markdown, describe what your charts and stats show.
- Explain insights clearly: what increases with what, and how strong is it?
- Discuss how context (like gender or time of day) changes interpretation.

6■■ Reflection

- What new relationship surprised you?
- Did the data confirm or challenge your intuition?
- What might you explore next (multivariate regression, outlier effects)?